1 predicates and quantifiers

propositional logic is simple, however limited. we can't use propositional logic to represent some classes of natural language statements such as this:

given that:
- all of my dogs like peanut butter, and
- kody is one of my dogs

can we draw the conclusion that kody likes peanut butter with propositional logic?

we also

the previous examples are called *syllogisms*.

1.1 predicates

predicate logic allows us to use *propositional functions* during our logical reasoning. for example:

$$P(x) \equiv x^3 > 0$$

in this propositional function, x is the variable, and $x^3 > 0$ is the predicate. note that propositional function $P(x)$ *has no truth value* unless it is evaluated for a given x or a set of xs. for example:

$$\begin{aligned} P(0) &\equiv \bot \\ P(23) &\equiv \top \\ P(-42) &\equiv \bot \end{aligned}$$

1.2 quantifiers

with *quantifiers*, we can make general statements that turn propositional functions into propositions. in english, we use quantifiers regularly:
- **all** students can ride the bus for free
- **many** people like chocolate
- i enjoy **some** types of tea
- **at least one** person will sleep through their final exam

quantifiers require us to define a *universe of discourse* (or a *domain*) in order for the quantification to make sense.

1.2.1 universal quantifier

a **universal quantifier** allows us to make statements about the *entire domain*. for example:

- **all** of my dogs like peanut butter
- **every** even integer is a multiple of two
- **for each** positive integer x, $2x > x$

in mathematical notation, we can express the universal quantification of $P(x)$ as $\forall x\, P(x)$.

for example, we can rewrite the sentence “the inequality $x + 1 < x$ holds for at least one integer” as a proposition with the universal quantifier:

1.2.2 existential quantifier

an **existential quantifier** allow us to make statements about *some objects* in a domain. for example:

- **some** elephants are scared of mice
- **there exist** integers a, b, and c such that the equality $a^2 + b^2 = c^2$ is true
- **there is at least one** person who did better than john on the midterm

given a propositional function $P(x)$, we express the existential quantification of $P(x)$ as $\exists x\, P(x)$.

1.3

2 activities

2.1 prove: $\exists x\,[P(x) \vee Q(x)] \equiv \exists x\, P(x) \vee \exists x Q(x)$